E/O/BP: Visualizing Knowledge Graphs

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Overview

- **Knowledge graphs** are an important component of modern AI systems.
- An intelligent agent such as the cognitive assistant being developed by AI-CARING will likely make use of knowledge graphs.
- Thus, to help people understand how AI-based cognitive assistants work, we should find ways to familiarize them with knowledge graphs.

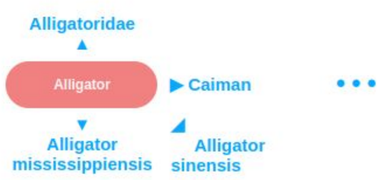
Previous Accomplishments

- Last year we created a knowledge graph with animal taxonomy information.
- The goal was to explore approaches to visualizing this information and for helping people to navigate through it.

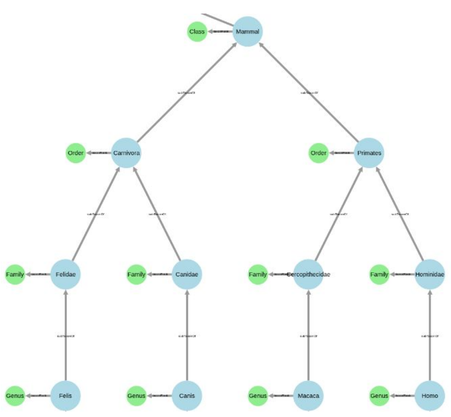
We developed four tools for visualization and navigation:

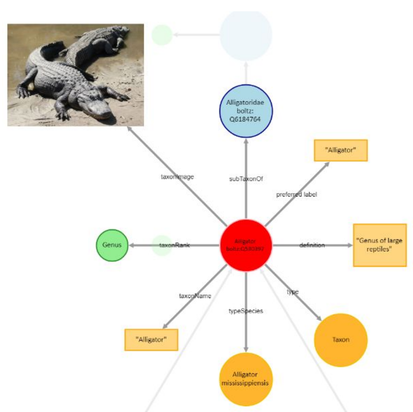
1. **Hierarchy Path Tool:** concisely shows the current node, the upward path to the root, and a selected downward path based on navigation history.

2. **Local Subgraph Navigator:** graphically depicts the current node along with its parent, siblings, and children. Allows navigation up, down, or sideways.

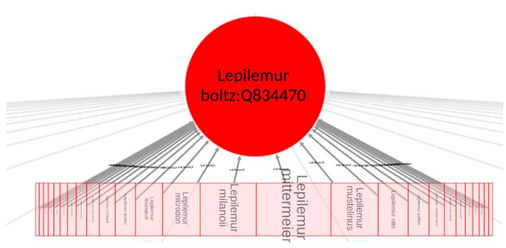


3. **Graph Visualization Tool:** Displays selected portions of the taxonomic hierarchy and provides for expanding selected nodes to show all their relations.





4. **Fish-Eye Node Selector:** provides spatially compact representation for nodes with many children to make child selection easier.

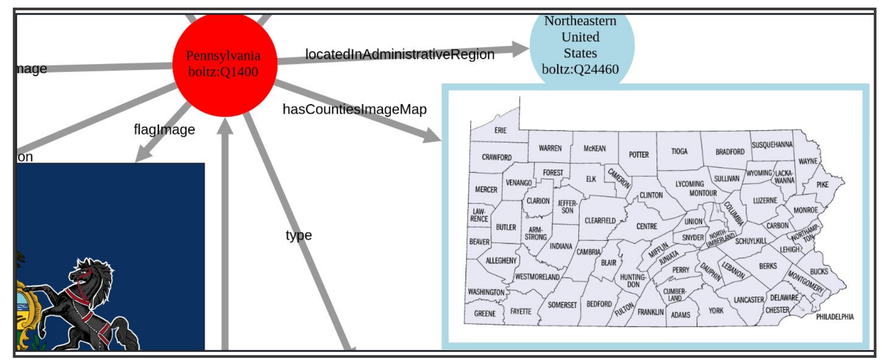


Approach

In the past year we've extended our approach to a second domain: geographic regions such as cities, counties, and states.

We reimplemented the previous tools for this new domain, which also has a hierarchical organization but is complicated by many level-skipping relations, e.g., Pittsburgh is in Allegheny County but also in Pennsylvania and the Northeastern United States.

We created a new graphical tool, **Image Map Selector**, as a graphical approach to selecting children of a geographic region node.



Click on a county in the image map to display that node as a child of Pennsylvania.

Browsing behavior for geographic regions is different than for the animal taxonomy because there can be many paths to a concept. We have developed a new display convention to maintain an ordered display as users move between levels.

Our next step is to develop generic, parameterized versions of our navigation and display tools that can apply to other domains besides taxonomy and geography. Promising domains include corporations (already in progress), consumer products, and medical information.

Results So Far

- We have a partial implementation of the generic navigation and display tools adapted to the geographic region domain.
- We have generalized the Image Map Selector to pull image region descriptions from the knowledge graph, so nothing is hard-coded.
- We have developed sample Python code to query the knowledge graph. This will allow more advanced students to explore knowledge graph inference on their own. Using our "best practices" demonstration knowledge graph is easier than querying WikiData.
- Demo links and sample code have been made available at https://www.cs.cmu.edu/~dst/KnowledgeGraphDemo

Future Work

- Complete the parameterized generic versions of our navigation and display tools and redo the animal taxonomy demo to use them.
- Continue work in progress on displaying ontology information (e.g., information about taxonomic rankings, or types of geographic regions) using new graphical conventions.
- Develop additional graphical tools for display of complex information, similar to fish-eye and image map tools. Example: scrollable menus for properties that are lists of concepts.
- Develop additional demo programs and exercises for students to explore knowledge graph inference using Python and SPARQL.
- Closer integration with AI-CARING concerns by adding new domains relevant to patients receiving care.

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